



Hospital Management System Using MERN Stack

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Introduction

Traditional hospital management which is done over the phone and records in paper format have lot of inefficiencies and inaccurate appointment scheduling. This project aims to design a user friendly application for patients and doctors to perform medical appointment scheduling, generating prescriptions and handling inventory using MERN Stack technology.

Motivation

To enhance patient experience and provide easy access to appointment scheduling, medicines and to reduce human errors in the physical process. Also, to access real time data for faster decision making.

SCOPE of the Project

- The HMS is a full stack web application which is designed to manage appointment scheduling and inventory. The 3 different users – admin, doctor and patient have different functionalities respectively. This application is scalable and allows the project to add any features in the future.
- It reduces physical work and paper records and enhances quality of medical services by storing the patient information online which is available anytime. This is a centralized platform and stores all data in MongoDB database.

Methodology

The main technologies which are used for this project is MERN Stack. Let’s get the brief overview of the frontend, backend and the database used to implement the application. The following tools and technologies have been used:

1. Front End

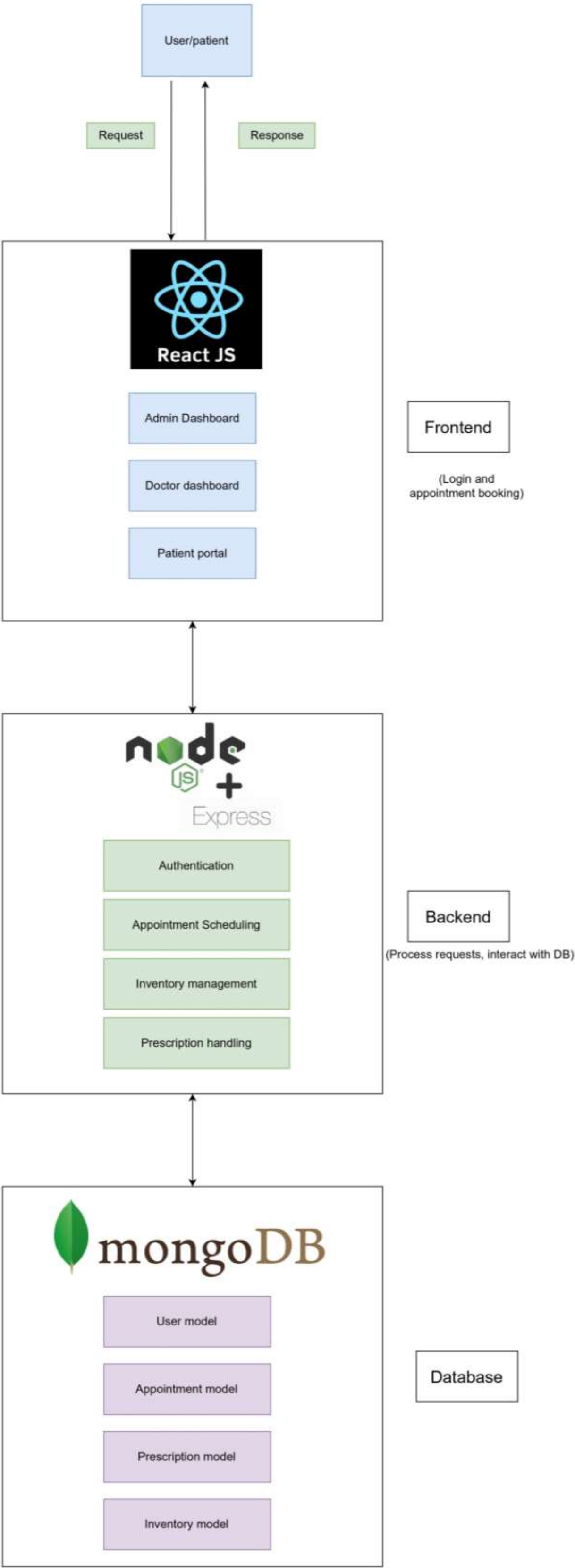
There are 3 different logins – Admin, Doctor and Patient. React JS project is classified into client and server. React is used for client. To install react project, place this command “npx create-react-app.” in the VS code. The node modules are installed and we are ready to perform frontend logic. The other packages required for the implementation can be installed accordingly by “npm install” command.

2. Back End

Node JS is installed in the VS Code by using npm install or directly installing latest version from the Google. Express JS is a Node JS framework used to build the application offering middleware, routing, etc. The backend is divided into 4 parts mainly. They are - Controllers, models, middlewares, and routes.

3. Database

For database, I used MongoDB. Create a MongoDB account using and select the dropdown menu to create a new project. It ensures to store the user credentials and their information.

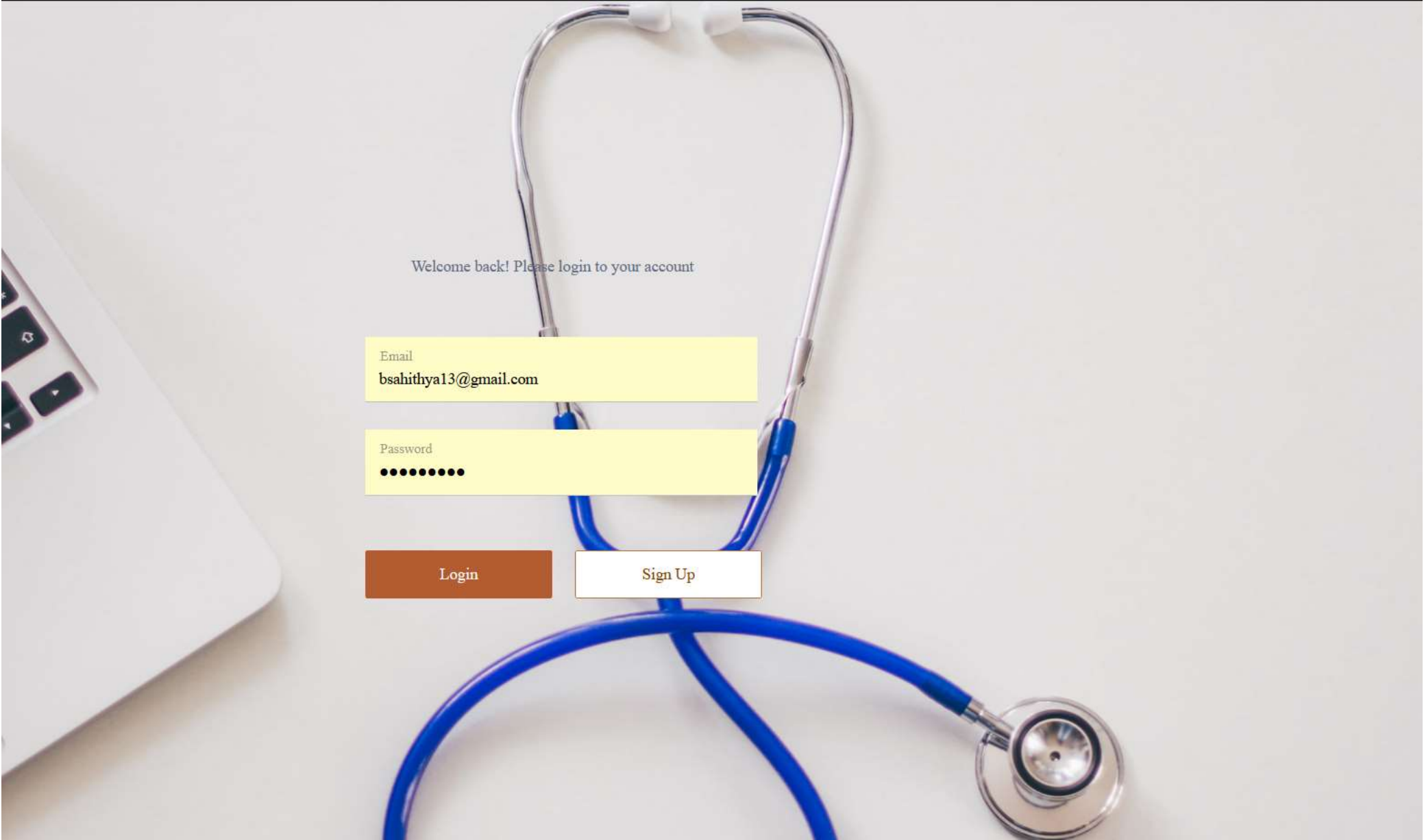


System Architecture of Hospital Management System

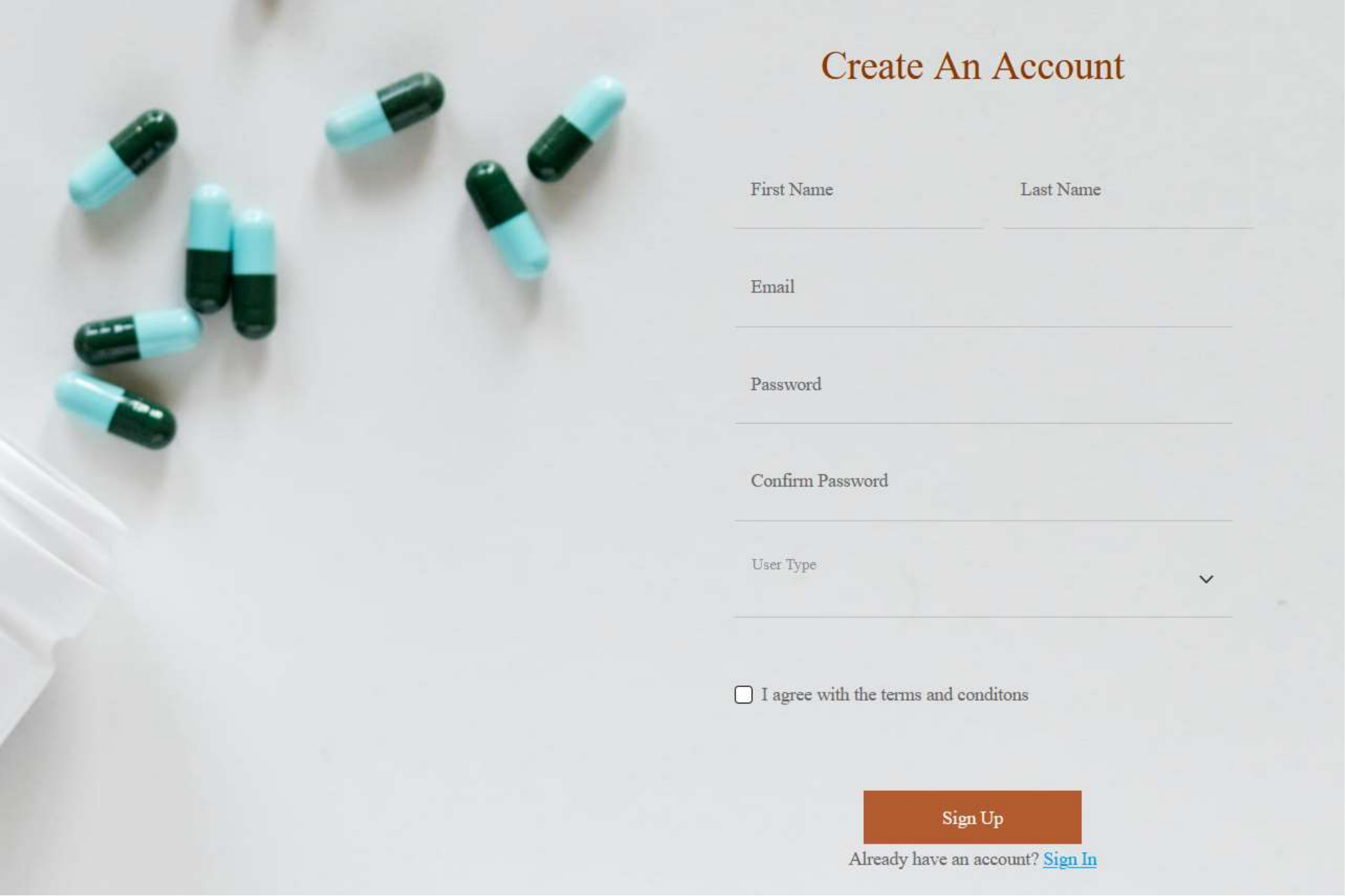
Results

The results of this project include:

- All the users – doctors, admins, and patients have a different dashboard based on the permissions and each has different benefits. A sign up page is created for new user. After a successful registration, user is ready to login to the website.
- After successful authentication, user can view the available doctors and their schedule. Users can select the doctor of their choice and book the appointment.
- Patient’s dashboard has the facility to check the schedule and available slots of the doctor to schedule an appointment. Admins can add, update and delete the user details which provide the authority of performing CRUD operation to the website.
- User profile is categorized as 3 profiles – Admin, Doctor and patient. Admin is the one who operate the prescriptions, inventory, and manage user details. Doctor can select their free time to check the patient and add the prescription. Patient can book the appointment and view the prescription.



Login Page



Signup page

Conclusion

The Hospital Management System implemented with the MERN stack effectively solves major issues such as appointment scheduling, patient record management, prescription management, and medicine inventory management. Automating repetitive tasks and providing role-based access, the system improves the efficiency, accuracy, and accessibility of healthcare facilities. It offers an easy-to-use interface for admins, doctors, and patients, making the experience smooth across all modules. The system helps in improved organization of data, less administrative burden, and enhanced patient care through prompt access to medical information. In the future, we can develop online payment module for appointment scheduling and inventory.

References

Odeh, A., Abdelhadi, R., & Odeh, H. (2019, December). Medical patient appointments management using smart software system in UAE. In 2019 International Arab Conference on Information Technology (ACIT) (pp. 97-101). IEEE.